Type MS · Zeitgebers, light and Siffre's cave study

Answer 1 of 9 · Question 33 part whole · 4 marks

2024 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Question number	Answer	Mark
33	**One** mark for identification of each way (maximum **two** marks). **One** mark for justification which must be linked to the way identified (maximum **two** marks). For example: • Leonardo may not know what day it is or when to sleep because of the lack of sunlight acting as an external cue to help him maintain his 24-hour sleep wake cycle (1). Siffre (1975) believed it was mid-July when he finished his study, but it was August, which shows how cues are important for maintaining bodily rhythms (1). • Being unable to sew buttons on a shirt shows that Leonardo began to struggle with his manual dexterity finding it hard to coordinate fine motor skills for sewing (1). Siffre (1975) highlighted that the effects of isolation from external cues could result in deterioration of physical dexterity (1). Accept any other appropriate response.	**(4)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 1 of 9 · Question 33 part whole · 4 marks

2024 JUN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Type MS · Zeitgebers, light and Siffre's cave study

Answer 2 of 9 · Question 34 part whole · 4 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Question number	Answer	Mark
34	**One** mark for each identification of each reason (maximum **two** marks). **One** mark for each justification, which must be linked to the reason identified (maximum **two** marks). For example: • The dark curtain will block light during the daytime which will be waking Nick up and preventing him from sleeping (1) because daylight is an external zeitgeber regulating the sleep-wake cycle making Nick feel awake (1). • The use of a special lamp in the evening will help Nick feel awake as it replicates daylight (1). This reduces the influence of the darkness on Nick's circadian rhythm so that he will not feel sleepy at night (1). Accept any other appropriate response.	**(4)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 2 of 9 · Question 34 part whole · 4 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract

Type MS · Zeitgebers, light and Siffre's cave study

Answer 3 of 9 · Question 27 part whole · 1 mark

2020 NOV · Paper 02 · find part whole in this mark scheme extract

Question number	Answer	Mark
27	B – memory tests **The only correct answer is B** A is incorrect as this was not a measure used C is incorrect as this was not a measure used D is incorrect as this was not a measure used	**(1)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 4 of 9 · Question 35 part whole · 9 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

Question number	Indicative content	Mark
35	AO1 = 3 marks; AO2 = 3 marks; AO3 = 3 marks Markers must apply the descriptors in line with the general marking guidance (on pages 2–3). Due to the intrinsic links between the skills, if a response evidences performance against only one trait/skill it will be for demonstrating knowledge and understanding. An answer displaying qualities of AO1 only, cannot be awarded more than the top of Level 1, no matter how strong the performance is in AO1. AO1 • Siffre (1975) spent six months in a cave where there was no external cue of daylight to indicate when it was night or day. • His sleep-wake cycle ranged from 18 to 51¼ hours, although each cycle felt like a single day. • Several psychological and physiological deteriorations were recorded, such as poor memory, low mood and poor dexterity. AO2 • When the communities are in darkness for six months, this is the same period of time Siffre spent in the cave. • People in the regions of the world in darkness may experience extended periods of being awake or asleep because of the lack of zeitgebers to indicate day and night. • The communities may have high levels of seasonal affective disorder in the dark winter months due to the lack of natural light. AO3 • The findings of the study could realistically represent the experiences of the people in the communities as they are in darkness for a similar period of time. • The study cannot explain what would happen during extended daylight months as Siffre (1975) only studied a period without natural light, so it may not explain the effect on the people in these regions. • Rosen et al. (1990) found a correlation between the more north a person lived and a higher prevalence of winter SAD, supporting the low mood found by Siffre (1975) which may also affect the people living in the regions that are dark.	**(9)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 4 of 9 · Question 35 part whole · 9 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding of a limited range of psychological ideas (AO1). • Provides little or no reference to relevant psychological ideas related to the context (AO2). • Limited attempt to deconstruct relevant psychological ideas. Limited consideration of supporting/refuting evidence, leading to generic judgements (AO3).
Level 2	4–6	• Demonstrates mostly accurate understanding of some relevant psychological ideas (AO1). • Provides some reference to relevant psychological ideas related to the context but this may be limited or lack relevance at times (AO2). • Deconstructs relevant psychological ideas using mostly logical chains of reasoning. Some consideration of supporting/refuting evidence, leading to a judgement (AO3).
Level 3	7–9	• Demonstrates accurate and thorough knowledge and understanding of relevant psychological ideas (AO1). • Provides sustained reference to relevant psychological ideas related to the context (AO2). • Deconstructs relevant psychological ideas using logical chains of reasoning. Sustained consideration of supporting/refuting evidence, showing an awareness of competing arguments, leading to a judgement (AO3).

Type MS · Zeitgebers, light and Siffre's cave study

Answer 4 of 9 · Question 35 part whole · 9 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Section A
Language, thought and communication – How do you communicate with others?

Type MS · Zeitgebers, light and Siffre's cave study

Answer 5 of 9 · Question 27 part whole · 1 mark

2022 JUN · Paper 02 · find part whole in this mark scheme extract

Question number	Answer	Mark
27	B - Approximately 18 to 51¾ hours **The only correct answer is B** A is incorrect as his sleep-wake cycles range was longer C is incorrect as his sleep-wake cycles range was longer D is incorrect as his sleep-wake cycles range was longer	**(1)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 6 of 9 · Question 29 part whole · 2 marks

2022 JUN · Paper 02 · find part whole in this mark scheme extract

Question number	Answer	Mark
29	Up to **two** marks for understanding the effect of light For example: • Light is an exogenous zeitgeber that is registered by the retina in the eye (1) that helps signal when it is time to sleep or wake up (1). Accept any other appropriate response.	**(2)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 7 of 9 · Question 31 part whole · 2 marks

2023 JUN · Paper 02 · find part whole in this mark scheme extract

Question number	Answer	Mark
31	**One** mark for identification of a reason. **One** mark for reasoning/justification through a study. For example: • Chris was in a dark room with no natural light cue to indicate whether it was day or night which could have affected his sleep-wake cycle and confused him about how long he was in the underground bunker (1). Siffre (1975) found that during his time in the cave he lost track of his sleep-wake cycle and even when he had extended cycles of over 48 hours, they still only felt like a day to him (1). Accept any other appropriate response.	**(2)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 8 of 9 · Question 33 part whole · 9 marks

2023 JUN · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

Question number	Indicative content	Mark
33	AO1 = 3 marks; AO2 = 3 marks; AO3 = 3 marks Markers must apply the descriptors in line with the general marking guidance (on pages 2–3). Due to the intrinsic links between the skills, if a response evidences performance against only one trait/skill it will be for demonstrating knowledge and understanding. An answer displaying qualities of AO1 only, cannot be awarded more than the top of Level 1, no matter how strong the performance is in AO1. AO1 • External influences on sleep are exogenous pacemakers that can help regulate the internal body clock, such as light or social cues. • Light is a zeitgeber that helps trigger the suprachiasmatic nucleus (SCN) to send messages to the pineal gland. • Mealtimes and activities are social cues that can also help regulate biological rhythms. AO2 • Artez may be feeling tired at work because it is dark, and his body clock would be expecting to sleep during the night-time due to there being no natural light. • The use of bright lighting when Artez wakes up would help trigger the SCN to send a message to the pineal gland to stop secreting melatonin so Artez would feel more awake. • Artez may feel less tired if he has large breakfast when he wakes as his body will be metabolising energy from the food at the right times for his 24-hour cycle. AO3 • Evidence from Siffre (1975) suggests that zeitgebers have an influential role as exogenous pacemakers in setting the internal body clock for the sleep-wake cycle, so his colleagues' advice should help Artez with his sleeping. • Skene and Arendt (2007) showed that blind people with some light perception have normal circadian rhythms but those without light perception show abnormal circadian rhythms, so the bright light may help Artez wake up. • Mealtimes and activities may be less successful than light as they do not directly trigger the SCN to message the pineal gland, therefore this advice may not help Artez with his sleep-wake cycle or resolve his tiredness at work.	**(9)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 8 of 9 · Question 33 part whole · 9 marks

2023 JUN · Paper 02 · extract 2 of 3 · find part whole in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–3	• Demonstrates isolated elements of knowledge and understanding of a limited range of psychological ideas (AO1). • Provides little or no reference to relevant psychological ideas related to the context (AO2). • Limited attempt to deconstruct relevant psychological ideas. Limited consideration of supporting/refuting evidence, leading to generic judgements (AO3).
Level 2	4–6	• Demonstrates mostly accurate understanding of some relevant psychological ideas (AO1). • Provides some reference to relevant psychological ideas related to the context but this may be limited or lack relevance at times (AO2). • Deconstructs relevant psychological ideas using mostly logical chains of reasoning. Some consideration of supporting/refuting evidence, leading to a judgement (AO3).
Level 3	7–9	• Demonstrates accurate and thorough knowledge and understanding of relevant psychological ideas (AO1). • Provides sustained reference to relevant psychological ideas related to the context (AO2). • Deconstructs relevant psychological ideas using logical chains of reasoning. Sustained consideration of supporting/refuting evidence, showing an awareness of competing arguments, leading to a judgement (AO3).

Type MS · Zeitgebers, light and Siffre's cave study

Answer 8 of 9 · Question 33 part whole · 9 marks

2023 JUN · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Section F
Language, thought and communication – How do you communicate with others?

Type MS · Zeitgebers, light and Siffre's cave study

Answer 9 of 9 · Question 31 part whole · 2 marks

2024 JUN · Paper 02 · extract 1 of 2 · find part whole in this mark scheme extract

Question number	Answer	Mark
31	**One** mark for accurate understanding of light. **One** mark for exemplification of how light can be used to explain the scenario. For example: • Light and dark act as zeitgebers that signal to the body when it is time to wake up or go to sleep (1) so turning the light off creates the darkness so Tallulah's body will begin to prepare for sleep at the time she wants to sleep (1). Accept any other appropriate response.	**(2)**

Type MS · Zeitgebers, light and Siffre's cave study

Answer 9 of 9 · Question 31 part whole · 2 marks

2024 JUN · Paper 02 · extract 2 of 2 · find part whole in this mark scheme extract